Second part: Open book

Total score: 10 pts.

You are warmly invited to be tidy and concise. Messy papers will be penalized.

5. An international company has a set of working places $O = \{1, \ldots, n\}$ in the UK and a set of other working places $D = \{1, \ldots, m\}$ all over the world. Due to the brexit, the workers in the UK must be transferred abroad. Let a_i be the number of workers to be transferred from $i \in O$ and let b_j be the maximum number of workers that working place $j \in D$ can accept.

   For each worker of place i moved to place j, the company has a cost of c_{ij}. Since the capacity of the destination places may not be sufficient to host all workers, or the cost for moving them may be too high, the company can decide to incentivize the severance of some workers. Let s_i the cost for firing a worker of place i.

   Formulate the problem of determining the transfer plan of workers that comply with the availability of positions in the host working places and minimize the total cost. The model must be linear.

   Hint: all workers are equivalent for the company, thus it is not necessary to introduce a variable for each worker.

   Variant: Let us assume that the company made an agreement with the union organizations to limit the overall number of fired employees to a maximum value of F_i. in each place $i \in O$ However, the agreement specifies also that if the limit of F_i fired employees is exceeded, the company has to pay a penalty fee of P_i to the union organization in place i. Introduce these explicit constraints and modify the objective function, specifying whether additional variables are needed and their meaning.

6. Consider a set of ingredients $I = \{1, \ldots, n\}$. Let $R = \{1, \ldots, m\}$ be the set of recipes contained in a cookbook. Every recipe $j \in R$ needs a subset of ingredients $S_j \subset I$, and the available quantity of each ingredient is such that it can be used in at most one recipe. A cook has to decide which recipes to prepare in such a way that each ingredient is used at most once, and the number of recipes is maximized. Describe a possible greedy algorithm for the problem, specifying the functions Best and Ind.

   Let us suppose that associated with each recipe $j \in R$ there is a revenue r_j, how does the function Best change?